Change of the Digested Sludge Quality at Microaerobic Digestion

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ABSTRACT: The implementation of microaerobic conditions often improves the results of anaerobic digestion. The study was focussed on comparison of digested sludge quality after shift from strictly anaerobic to microaerobic sludge digestion. The compared parameters were: chemical composition of sludge, specific activity of methanogenic bacteria, foaming potential and dewaterability of sludge. In most parameters the quality of sludge digested at microaerobic and anaerobic conditions is comparable often the quality of microaerobic sludge is better. From the operational point of view it is very important that foaming of microaerobic digester was lower and the dewaterability was better for microaerobic sludge.

INTRODUCTION

THE quality of digested sludge is influenced significantly by specific conditions of digestion process. Some studies comparing the quality of sludge in relation to operational conditions or sludge pretreatment methods were published [1, 2]. Little is known about influence microaerobic conditions or increase of oxidation-reduction potential (ORP) on quality of digested sludge.

It has been proved in different studies that the use of microaerobic conditions in anaerobic digestion can be beneficial. The application of microaerobic technology is an efficient method of hydrogen sulphide removal from biogas [3, 4, 5, 6] and sulphide toxicity suppression [7, 8]. The implementation of microaerobic conditions has in many cases improved the efficiency of anaerobic digestion and enhanced hydrolysis and biodegradability of some organic compounds [9, 10].

In a mixed culture, even strict anaerobes can survive without any inhibition, if facultative microorganisms are able to consume the present oxygen quickly and fully. It was shown that the presence of limited amount of oxygen in digester does not destroy the digestion process even in the system where the oxygen is not consumed by sulphide oxidation [11, 12].

In this paper, microaerobic sludge means the sludge cultivated in anaerobic system with limited (trace) oxygen consumption. With respect to the ORP, the microaerobic system can be marked generally as a system in which micro-consumption of oxygen causes a limited ORP increase [13].

Microaerobic conditions are obtained by dosing of a limited amount of air (oxygen can be used as well) into anaerobic reactor. There are two basic configurations of microaerobic technology. The air can be pumped into the gas space of the digester or into the mixed liquor suspension. In the first alternative, the risk of biogas pipe clogging by sulphur is higher; at the second alternative a bigger surplus of oxygen is needed. Moreover because of closer contact of sludge with oxygen the likelihood of the bacterial consortia affection is more significant in the second case.

There were published studies comparing the specific bacterial activity after a shift from strictly anaerobic to microaerobic sludge digestion. Tang [14] reported that microaeration has no obvious effect on the phylogenetic diversity of microorganisms. However the results indicated that ratio between hydrogenotrophic and acetoclastic methanogens changed due to microaeration. Zitomer and Shrout [15] reported that the methanogenic activity can sometimes be even higher under microaerobic conditions in comparison with strictly anaerobic system.

Less attention has been paid till now to comparison of digested sludge quality after anaerobic and microaerobic digestion. It is therefore the main aim of presented paper. The present paper aims to evaluate the changes in chemical composition, foaming potential, dewaterability and methanogenic activity after switch from fully anaerobic to microaerobic conditions.

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MATERIAL AND METHODS

The experiments were carried out in two continuously stirred lab-scale reactors with 10 L of working volume. Reactors were operated at mesophilic temperature. Total duration of parallel operation of both reactors was 208 days.

The start-up of both reactors was conducted in anaerobic conditions. After start-up, the operation of the first reactor was changed to microaerobic. To the bottom of one of them was continuously dosed air. Second reactor remained anaerobic and served as referential. Waste activated sludge of municipal wastewater treatment plant was used as the substrate for reactors; the sulphur content was increased by natrium sulphate addition. The average sludge composition was following: total suspended solids (TSS) 32.8 g/L, volatile suspended solids (VSS) 20.1 g/L, pH 7.84.

The reaction mixture was kept homogeneous by mechanical mixing and operational temperature was kept at 40 ± 1°C. The volumetric loading rates of digesters were 2.0 g/L.d and 0.15 g/L.d for chemical oxygen demand (COD) and SO_4^{2-} respectively. The retention time in both reactors was on average 30 days. To reach the microaerobic conditions and oxidize the sulphide produced by sulphate reducing bacteria, one reactor was connected to an air supply by peristaltic pump with flow rate fixed to 1.7 L/d.

Sludge Composition

Analytical procedures were carried out according to Standard Methods for the Examination of Water and Wastewater [16], the biogas composition and volatile fatty acids were determined by gas chromatograph GC 8000Top equipped with a heat conductivity detector HWD 800 [17].

The elemental composition of sludge was assessed by X-ray fluorescence analysis, using the ARL 9400 XP sequential WD-XRF spectrometer. It is equipped with the Rh anode end-window X-ray tube type 4GN fitted with 75 μm Be window. All peak intensity data were collected by software WinXRF in vacuum.

Specific Activity of Methanogenic Bacteria

The assessment of the specific methanogenic activity (SMA) was carried out in serum bottles under mesophilic conditions without mixing using initial loading of 1 g COD/g VSS. All tests were carried out in three replicates. The volume of produced methane was monitored and maximum methane production rate calculated according to guidelines proposed by Angelidaki *et al.*, [18]. The production of biogas was measured volumetrically using water displacement method. The biogas composition was determined by a gas chromatography. The tests were performed with acetate as organic substrate and therefore the activity of acetoclastic methanogens was assessed.

Foaming Potential and Foam Stability

These parameters were assessed as additional characteristics of digested sludge, because presence of filamentous foam forming bacteria was elevated. It has been reported, that anaerobically treated excess activated sludge exhibits the tendency to cause foaming in digester. For the description and comparison of foam quantity and quality the "bubble test" has been developed [19]. The testing was based on the test described by Pagilla *et al.*, [20] and modified with respect to the character of anaerobic sludge. The final test was carried out by bubbling of 1 liter of sludge by nitrogen with the flow rate of 1 L/min in 2 L volumetric cylinders. The level of foamy sludge is recorded after 5 minutes of bubbling and the foaming potential (FP) is calculated from this value [Equation (1)]. 5 minutes after the stopping of the gas flow the level of foamy sludge is recorded again to calculate index of stability (IS), [Equation (2)]. FP describes the capability of the sludge to create foam. IS gives the information about stability of the foam created.

$$FP = \frac{V_5}{V_0} \tag{1}$$

where

$$IS = \frac{(V_{ST} - V_0)}{(V_5 - V_0)} \times 100 \tag{2}$$

V_0 = volume of sludge at the beginning of measuring (usually 1 L)
V_5 = volume of foamy sludge after 5 min of bubbling
V_{ST} = volume of foamy sludge 5 min after the gas flow stop

Microscopy

The microscopic analysis of sludge samples was performed according to Jenkins *et al.* [21]. The microscopic examination in wet mounts (Olympus BH2-RFCA—125×, 250× and phase contrast 1250× mag-

nification) was aimed at basic biomass characteristics (morphological properties, structure, presence and types of protozoa/metazoa and zoogloeal colonies). Gram and Neisser staining procedures were applied to determine the abundance and to identify filamentous microorganisms.

CST

Capillary suction time (CST) is a simple parameter and method used for characterization of sludge dewaterability. Relationship between dewaterability measured by CST and a number of sludge properties has been shown in the past and it was found that the CST test is accurate, if the product of solid concentration and specific resistance to filtration is of interest [22]. The original circular setup introduced by Gale and Baskerville [23] and Whatman-17 filter paper were used for measurement.

RESULTS AND DISCUSSION

The main results of anaerobic digestion process for both reactors are shown in Table 1. It is evident that despite air dosing, the specific methane production in the microaerobic digester was higher. Significantly better quality of sludge liquor as regards soluble COD was proved. Rest of nitrogen from dosed air caused lower methane concentration in biogas of microaerobic digester, however if we balance the methane production the result of microaerobic digester is slightly better, due to suppressed inhibition by sulphides.

Table 1 shows that also variation in basic parameters of the sludges is small. Bigger difference can be found in sludge liquor composition (soluble COD). Higher concentration of total sulphide in anaerobic sludge is not in contradiction with data about sulphur in Table 2 and indicates the oxidation of sulphide.

In the evaluated period the hydrogensulphide concentration in biogas produced in the anaerobic reactor was high (almost 7.3 g/m^3), therefore estimated H_2S removal efficiency by microaerobic digester is above 99% confirming results previously reported [11, 24].

Sludge Composition

Small differences were found in elemental composition of digested sludge from microaerobic and anaerobic reactor (Table 2). Expected increase of sulphur content due to precipitation of elemental S in sludge was confirmed, however this increase was not significant because of continuous removal of sludge (and sulphur) from digester. High Fe content is caused by Fe3+ addition in wastewater treatment technology where the treated activated sludge originates from.

Specific Activity of Methanogenic Bacteria

Activity of methanogenic bacteria was periodically monitored during the digesters operation. The results of activity tests are shown in Figure 1. Immediately after the start of microaeration, decrease (16%) of specific methanogenic activity (SMA) was observed. When sulphate dosing was initiated, methanogenic activity of both anaerobic and microaerobic biomass increased, possibly because of previous deficit of sulphur. During further operation, the SMA dropped again in both reactors, however, the decrease of activity of microaerobic biomass was smaller and therefore its activity was higher in comparison with the anaerobic biomass. This activity development was mainly affected by dissolved sulphide concentration probably—Figure

Table 1. Comparison of Digestion Results and Sludge Quality (Average Value and Standard Deviation).

Parameter	Microaerobic Digester	Anaerobic Digester
soluble COD (g/L)	3.44 ± 0.23	4.80 ± 0.31
specific methane production (L/kg VSS_{added})	330 ± 1.5	305 ± 2.1
hydrogensulphide in biogas (mg/m^3)	47 ± 14.4	7298 ± 489
TSS (g/L)	27.8 ± 0.93	27.2 ± 0.42
VSS (g/L)	16.4 ± 0.40	16.1 ± 0.28
VSS/TSS (%)	58.98 ± 0.66	59.29 ± 0.59
COD (g/L)	32.5 ± 1.5	32.7 ± 1.5
Total Sulphide (mg/L)	615 ± 46	858 ± 43
ORPH (mV)	–278 ± 11	–309 ± 2.5

Table 2. Relative Elemental Composition of Digested Sludge from Microaerobic and Anaerobic Reactor (Major Elements of Mineral Fraction), (132nd day).

Element	Microaerobic Sludge (%)	Anaerobic Sludge (%)	Ratio Anaerobic/ Aerobic
Fe	28.7 ± 0.1	29.2 ± 0.1	0.983
P	17.3 ± 0.1	17.1 ± 0.1	1.012
Ca	11.9 ± 0.09	11.9 ± 0.09	1.000
S	9.92 ± 0.09	9.68 ± 0.09	1.025
Na	9.58 ± 0.09	9.66 ± 0.09	0.992
Al	8.48 ± 0.09	8.30 ± 0.09	1.022
Si	5.60 ± 0.07	5.43 ± 0.07	1.031
K	3.67 ± 0.06	3.80 ± 0.06	0.966
Mg	1.43 ± 0.04	1.44 ± 0.04	0.993

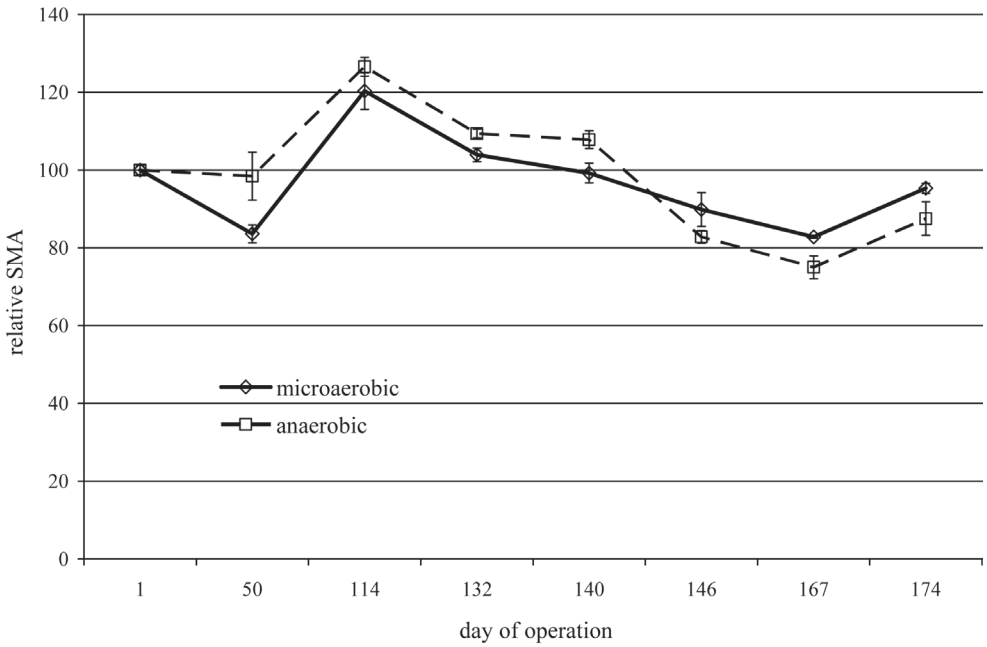


Figure 1. *The course of methanogenic bacteria activity.*

2. Sulphide concentration was gradually rising in both reactors; however, the concentration in microaerobic reactor was always significantly lower. IC_{50} (median inhibition values) reported for methanogenic activity range between 30 and 250 mg/L [25], however free H_2S is most toxic form and therefore also pH value is important. At actual pH of digesters (8.15 respectively 8.13) the maximum concentration of free H_2S was 14 and 24 mg/L for microaerobic and anaerobic digester respectively.

Foaming Problems

The waste activated sludge treated contained in some periods increased amount of filamentous bacteria and therefore the risk of digester foaming was high. A fear of stimulation of aerobic bacteria of activated sludge treated in digester existed at the start of microaerobic digester operation. The presence of oxygen could help to the survival of aerobic filamentous bacteria in microaerobic digester and consequently to stimulate the

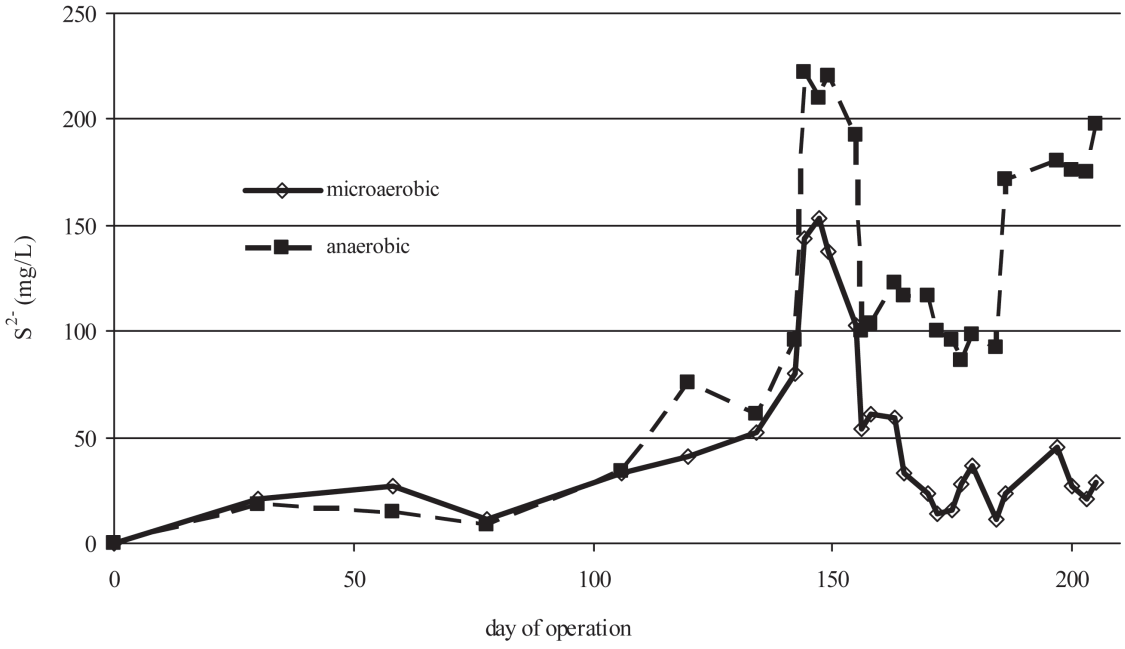


Figure 2. *The course of dissolved sulphide concentration.*

foam production. The results overcame the fear of behavior of aerobic filamentous bacteria in microaerobic conditions.

Especially during period with high foaming potential, the difference between microaerobic and anaerobic digester was high. The maximum foaming potential FP was 3.2 and 2.6 for the anaerobic and microaerobic reactor, respectively. In addition, the foam rising in anaerobic reactor was much more stable in comparison to the microaerobic reactor. The index of stability IS was 96% and 53% in the anaerobic and microaerobic reactor, respectively.

Better degradation of filaments of activated sludge (with dominance of *Microthrix parvicella*) during microaerobic digestion can be illustrated by microscopic pictures of digested sludge in Figure 3, where the fragments of filaments are in anaerobic sludge longer and more frequent and more compact.

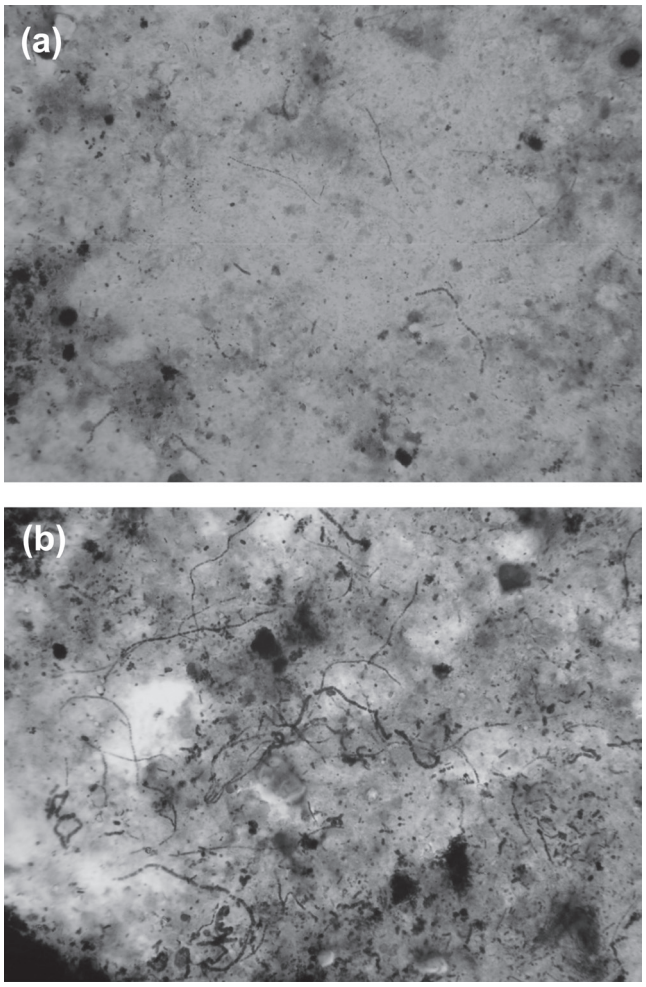


Figure 3. *Microscopic pictures of digested sludge; (a) microaerobic and (b) anaerobic, showing different abundance and morphology of filaments (Gram staining, direct light, enlargement 1250x magnification).*

Table 3. Comparison of CST Value (in seconds) for Digested Sludge from Microaerobic and Anaerobic Reactor.

Parameter	Average, std. dev.	Minimum	Maximum
Microaerobic digester	1484 ± 366	1176	2116
Anaerobic digester	2052 ± 672	1499	3148

Evaluation of Dewaterability by CST

Dewaterability was characterized by periodic CST measurement of sludge samples from both digesters. The results are shown in Table 3 and indicate that the dewaterability of microaerobic sludge is slightly better, however from the difference is not statistically significant. The CST values are relatively high due to specific character of treated sludge and due to high TSS and dissolved salts concentrations 27–30 g/L and 4.5–5.5 g/L respectively. Similar results were found by Ayol [26] and Wang and Dentel [27].

CONCLUSIONS

Based on the presented results of the long term monitoring of sludge quality of anaerobic and microaerobic digesters we can draw following conclusions:

- The quality of sludge in both reactors is similar, only small differences were found in composition and activities.
- Slightly higher content of total S and lower content of S-S^{2-} in microaerobic sludge indicate the accumulation of elemental S.
- Methanogenic activity slightly decreased due to air dosing in the digester, however, the methanogenic activity of microaerobic biomass was higher than that of strictly anaerobic biomass at higher sulphide concentrations.
- Composition of sludge liquor differs mainly in soluble COD concentration which is significantly lower at microaerobic digestion.
- The sludge digested in microaerobic conditions manifested lower foaming potential and foam stability, dewaterability characterized by CST was not significantly better for microaerobic sludge.

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